

Cài NGINX Ingress Controller

Thêm Helm repository, xác nhận chart

```
master@master:~$ helm repo add ingress-nginx https://kubernetes.github.io/ingress-nginx
"ingress-nginx" has been added to your repositories
master@master:~$ helm repo update
Hang tight while we grab the latest from your chart repositories...
...Successfully got an update from the "ingress-nginx" chart repository
Update Complete. ✨Happy Helming!✨
master@master:~$ helm search repo ingress-nginx/ingress-nginx
NAME                                CHART VERSION  APP VERSION     DESCRIPTION
ingress-nginx/ingress-nginx        4.15.1         1.15.1          Ingress controller for Kubernetes using NGINX a...
```

Cài Ingress Controller

```
master@master:~$ helm install ingress-nginx ingress-nginx/ingress-nginx \
--namespace ingress-nginx \
--create-namespace \
--set controller.service.type=NodePort \
--set controller.service.nodePorts.http=30080 \
--set controller.service.nodePorts.https=30443
NAME: ingress-nginx
LAST DEPLOYED: Fri Sep 18 13:15:18 2026
NAMESPACE: ingress-nginx
STATUS: deployed
REVISION: 1
DESCRIPTION: Install complete
```

Kiểm tra

```
master@master:~$ kubectl get ingressclass
NAME      CONTROLLER          PARAMETERS  AGE
nginx     k8s.io/ingress-nginx <none>      33s
master@master:~$ kubectl wait \
--namespace ingress-nginx \
--for=condition=Ready pod \
--selector=app.kubernetes.io/component=controller \
--timeout=120s
pod/ingress-nginx-controller-6c7cd85885-9vsp7 condition met
master@master:~$ kubectl get pods -n ingress-nginx
NAME                                READY   STATUS    RESTARTS   AGE
ingress-nginx-controller-6c7cd85885-9vsp7  1/1     Running   0           61s
```

Tạo name space

```
master@master:~$ kubectl config view --minify | grep namespace
namespace: ingress-lab
```

Tạo app1,2.yaml

```
master@master:~$ vim app-v1.yaml
master@master:~$ kubectl apply -f app-v1.yaml
deployment.apps/app-v1 created
service/app-v1-svc created
master@master:~$ vim app-v2.yaml
master@master:~$ kubectl apply -f app-v2.yaml
deployment.apps/app-v2 created
service/app-v2-svc created
```

Kiểm tra pods và service ip cluster

```

master@master:~$ kubectl get deployments
kubectl get pods -o wide
kubectl get svc
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
app-v1    2/2     2            2           24s
app-v2    2/2     2            2           5s
NAME      READY   STATUS    RESTARTS   AGE   IP             NODE      NOMINATED NODE
READINESS GATES
app-v1-89fb44f46-tg59t  1/1     Running    0          24s   10.233.102.139  node1    <none>
app-v1-89fb44f46-zvwt4  1/1     Running    0          24s   10.233.75.9     node2    <none>
app-v2-6c5cbf8677-cdrdr 1/1     Running    0          5s    10.233.102.140  node1    <none>
app-v2-6c5cbf8677-xf99t 1/1     Running    0          5s    10.233.75.10    node2    <none>
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
app-v1-svc ClusterIP    10.233.63.55 <none>        80/TCP    24s
app-v2-svc ClusterIP    10.233.1.134 <none>        80/TCP    5s

```

Kiểm tra app

```

master@master:~$ curl http://10.233.63.55
curl http://10.233.1.134
Hello from APP V1
Hello from APP V2

```

Kiểm tra endpoint

```

master@master:~$ kubectl get endpoints app-v1-svc app-v2-svc
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
NAME      ENDPOINTS                                     AGE
app-v1-svc 10.233.102.139:5678,10.233.75.9:5678        14m
app-v2-svc 10.233.102.140:5678,10.233.75.10:5678        14m

```

Ingress Host-based Routing

Tạo ingress.yaml

Nginx controller nhận thành công

```

master@master:~$ vim ingress.yaml
master@master:~$ kubectl apply -f ingress.yaml
ingress.networking.k8s.io/lab-ingress created
master@master:~$ kubectl get ingress
kubectl describe ingress lab-ingress
NAME      CLASS   HOSTS                                     ADDRESS   PORTS   AGE
lab-ingress  nginx  shop.lab.local,api.lab.local            80       5s
Name:      lab-ingress
Labels:    <none>
Namespace: ingress-lab
Address:
Ingress Class:  nginx
Default backend: <default>
Rules:
  Host      Path  Backends
  ----    -
  shop.lab.local  /    app-v1-svc:80 (10.233.102.139:5678,10.233.75.9:5678)
  api.lab.local   /    app-v2-svc:80 (10.233.75.10:5678,10.233.102.140:5678)
Annotations:  <none>
Events:
  Type    Reason    Age   From          Message
  ----    -
  Normal  Sync      5s    nginx-ingress-controller  Scheduled for sync

```

test Host-based Routing

```
master@master:~$ kubectl get nodes -o wide
NAME          STATUS    ROLES    AGE   VERSION   INTERNAL-IP   EXTERNAL-IP   OS-IMAGE
6.8.0-100-generic  Ready    control-plane  47h   v1.35.4   192.168.174.141 <none>        Ubuntu 24.04.5 LTS
node1          Ready    <none>      47h   v1.35.4   192.168.174.142 <none>        Ubuntu 24.04.4 LTS
6.8.0-139-generic  Ready    control-plane  47h   v1.35.4   192.168.174.143 <none>        Ubuntu 24.04.4 LTS
node2          Ready    <none>      47h   v1.35.4   192.168.174.143 <none>        Ubuntu 24.04.4 LTS
6.8.0-139-generic  Ready    control-plane  47h   v1.35.4   192.168.174.143 <none>        Ubuntu 24.04.4 LTS
master@master:~$ curl -H "Host: shop.lab.local" http://192.168.174.142:30080
Hello from APP V1
master@master:~$ curl -H "Host: api.lab.local" http://192.168.174.142:30080
Hello from APP V2
```

Cấu hình etc/hosts

Sudo vim /etc/hosts

Thêm vào file

192.168.174.142 shop.lab.local

192.168.174.142 api.lab.local

Kiểm tra rồi test

```
master@master:~$ getent hosts shop.lab.local
getent hosts api.lab.local
192.168.174.142 shop.lab.local
192.168.174.142 api.lab.local
master@master:~$ curl http://shop.lab.local:30080
curl http://api.lab.local:30080
Hello from APP V1
Hello from APP V2
```

TLS/HTTPS với self-signed certificate

tạo certificate dùng được cho cả hai hostname bằng SAN

```
master@master:~$ openssl req -x509 -nodes -days 365 \
  -newkey rsa:2048 \
  -keyout lab-tls.key \
  -out lab-tls.crt \
  -subj "/CN=shop.lab.local" \
  -addext "subjectAltName=DNS:shop.lab.local,DNS:api.lab.local"
```

Kiểm tra file và san

```
master@master:~$ ls -lh lab-tls.crt lab-tls.key
-rw-rw-r-- 1 master master 1.2K Sep 18 13:42 lab-tls.crt
-rw----- 1 master master 1.7K Sep 18 13:42 lab-tls.key
```

```
master@master:~$ openssl x509 -in lab-tls.crt -noout \
  -subject -issuer -dates -ext subjectAltName
subject=CN = shop.lab.local
issuer=CN = shop.lab.local
notBefore=Sep 18 13:42:33 2026 GMT
notAfter=Sep 18 13:42:33 2027 GMT
X509v3 Subject Alternative Name:
  DNS:shop.lab.local, DNS:api.lab.local
```

tạo Kubernetes TLS Secret

Tạo và kiểm tra secret: **Secret phải cùng namespace**

```
master@master:~$ kubectl create secret tls lab-tls-secret \
  --cert=lab-tls.crt \
  --key=lab-tls.key
secret/lab-tls-secret created
master@master:~$ kubectl get secret lab-tls-secret
NAME          TYPE          DATA  AGE
lab-tls-secret  kubernetes.io/tls  2      5s
master@master:~$ kubectl describe secret lab-tls-secret
Name:          lab-tls-secret
Namespace:     ingress-lab
Labels:        <none>
Annotations:   <none>

Type:  kubernetes.io/tls

Data
====
tls.crt:  1184 bytes
tls.key:  1704 bytes
```

cấu hình TLS cho Ingress

File ingress.yaml: chuyển cấu hình từ port 80 thành 80,443

```
master@master:~$ vim ingress.yaml
master@master:~$ kubectl apply -f ingress.yaml
ingress.networking.k8s.io/lab-ingress configured
master@master:~$ kubectl get ingress
kubectl describe ingress lab-ingress
NAME          CLASS  HOSTS          ADDRESS      PORTS      AGE
lab-ingress    nginx  shop.lab.local,api.lab.local  10.233.38.32  80, 443    12m
Name:          lab-ingress
Labels:        <none>
Namespace:     ingress-lab
Address:       10.233.38.32
Ingress Class: nginx
Default backend: <default>
TLS:
  lab-tls-secret terminates shop.lab.local,api.lab.local
Rules:
  Host          Path  Backends
  ----          -
  shop.lab.local  /    app-v1-svc:80 (10.233.102.139:5678,10.233.75.9:5678)
  api.lab.local  /    app-v2-svc:80 (10.233.75.10:5678,10.233.102.140:5678)
Annotations:   <none>
Events:
  Type    Reason    Age          From          Message
  ----    -
  Normal  Sync      3s (x3 over 12m)  nginx-ingress-controller  Scheduled for sync
```

Lưuồng hiện tại

Client

|
| HTTPS :30443



NGINX Ingress Controller

|

```

├── TLS termination
│   └── lab-tls-secret
├── shop.lab.local → app-v1-svc:80
└── api.lab.local → app-v2-svc:80

```

test HTTPS thực tế qua NodePort 30443

```

Normal Sync ... SS (X3 over 12m) nginx-ingress-controller Scheduled for
master@master:~$ curl https://shop.lab.local:30443
curl: (60) SSL certificate problem: self-signed certificate
More details here: https://curl.se/docs/sslcerts.html

curl failed to verify the legitimacy of the server and therefore could not
establish a secure connection to it. To learn more about this situation and
how to fix it, please visit the web page mentioned above.
master@master:~$ curl --cacert lab-tls.crt https://shop.lab.local:30443
Hello from APP V1
master@master:~$ curl --cacert lab-tls.crt https://api.lab.local:30443
Hello from APP V2

```

Cập nhật trust Certificates

```

master@master:~$ sudo cp lab-tls.crt /usr/local/share/ca-certificates/lab-tls.crt
master@master:~$ sudo update-ca-certificates
Updating certificates in /etc/ssl/certs...
rehash: warning: skipping ca-certificates.crt, it does not contain exactly one certificate or CRL
1 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
master@master:~$ curl https://shop.lab.local:30443
curl https://api.lab.local:30443
Hello from APP V1
Hello from APP V2

```

Deploy 1 cái web: ingress controller giúp nhiều app sài chung 1 port

DEPLOYMENT: cấu hình số pod..

Service: truy xuất port nào...

Ingress: quy định rules chạy service nào..

Path-based Routing

THÊM 1 VERSION CHO WEB

```

master@master:~$ kubectl describe ingress lab-ingress
Name: lab-ingress
Labels: <none>
Namespace: ingress-lab
Address: 10.233.38.32
Ingress Class: nginx
Default backend: <default>
TLS:
  lab-tls-secret terminates shop.lab.local,api.lab.local
Rules:
  Host            Path  Backends
  ----            -
  shop.lab.local  /v1   app-v1-svc:80 (10.233.102.139:5678,10.233.75.9:5678)
                  /v2   app-v2-svc:80 (10.233.75.10:5678,10.233.102.140:5678)
  api.lab.local   /     app-v2-svc:80 (10.233.75.10:5678,10.233.102.140:5678)

```

```
master@master:~$ curl https://shop.lab.local:30443/v1
curl https://shop.lab.local:30443/v2
Hello from APP V1
Hello from APP V2
master@master:~$ curl https://api.lab.local:30443/
Hello from APP V2
```

```
HTTPS :30443
|
├─ shop.lab.local/v1
|   └─ app-v1-svc → Hello from APP V1 ✓
|
├─ shop.lab.local/v2
|   └─ app-v2-svc → Hello from APP V2 ✓
|
└─ api.lab.local/
    └─ app-v2-svc → Hello from APP V2 ✓
```